

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Supplementary Winter 2023

Course: SY. B. Tech.

Branch: AI and Allied

Semester : IV

Subject Code & Name: BTAIPE405B & Image Processing and Computer Vision

Max Marks: 60

Date: 25/01/2024

Duration: 3.00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

- A) Differentiate Image Processing & Computer Vision. Explain the application of Image Processing and Computer Vision. (BT2/CO1) 6
- B) Draw and explain the block diagram of fundamental steps in Digital Image Processing. (BT1/CO1) 6
- C) Formulate and explain distance measures between two pixels. (BT2/CO1) 6

Q.2 Solve Any Two of the following.

- A) Define gray-level intensity transformation. Elaborate methods of gray level intensity transformation. (Linear, Logarithmic & Power law correction). (BT2/CO2) 6
- B) Suppose that a 3-bit image of size 64 x 64 pixels has the intensity distribution as shown in table 1. Get the histogram transformation function and make it out the image with the specified histogram as in table 2. Perform histogram matching. (BT4/CO2) 6

Table 1

r_k	0	1	2	3	4	5	6	7
n_k	760	1043	860	626	339	255	127	86

Table 2

z_k	0	1	2	3	4	5	6	7
PDF $P_z(z_q)$	0	0	0	0.15	0.20	0.30	0.20	0.15

- C) Using the listed smoothing filters, find the value of the pixel located at (2,2) in the given image (3 x 3 smoothing). (BT3/CO2) 6
1. Box filtering / Mean Filter
 2. Weighted Average filtering / Weighted Mean Filter
 3. Median filters
 4. Max filters
 5. Min filters

$$\text{Given Image} = \begin{bmatrix} 10 & 20 & 12 & 10 \\ 25 & 13 & 18 & 11 \\ 17 & 19 & 14 & 13 \\ 5 & 30 & 7 & 22 \end{bmatrix}$$

$$\text{Weight} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Q.3 Solve Any Two of the following.

- A) Define and explain Boundaries Extraction with suitable example. (BT2/CO3) 6
- B) Discuss the Thinning & Thickening algorithms in brief with examples. (BT2/CO3) 6
- C) Compute opening and closing of A & B. (BT4/CO1) 6

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \& \quad B = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Q.4 Solve Any Two of the following.

- A) Explain edge detection in detail. Also, define operators (Roberts, Prewitt, Sobel, and Laplacian). (BT1/CO4) 6
- B) Explain segmentation by region splitting and merging with example. (BT2/CO4) 6
- C) Write a note on Texture. (BT3/CO4) 6

Q.5 Solve Any Two of the following.

- A) Explain pattern classification by prototype matching. (BT2/CO5) 6
- B) What is Neural Network? Explain the Deep Learning. (BT2/CO5) 6
- C) Explain matching structural prototypes based on matching shape numbers. (BT2/CO5) 6

*** End ***